KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · SCIENCE

Earth's Moving Plates: Earthquakes, Volcanoes and Mountains

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 23):** Name the two main groups of decomposers. Then explain where the matter in a dead leaf goes when they break it down.

2. What do you think causes an earthquake? Write your best idea.

3. Name any volcano or mountain range you have heard of. Where is it?

4. Draw what you think is under the ground beneath a volcano. Label your drawing.

5. Predict: Do you think the continents have always been in the same places? Why or why not?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Plates in Motion

Use evidence from today's lesson. Write in complete sentences.

1. **Name the boundary** (divergent, convergent or transform): (a) the Andes rise as an ocean plate sinks under South America; (b) Iceland slowly splits; (c) land slides past land along the San Andreas fault.

2. **Read the data.** Alaska 1964 was magnitude 9.2; Mount St. Helens 1980 was magnitude 5.1. What is the difference? Which released more energy, and how do you know?

3. **Two-step problem.** The San Andreas fault moves about 5 centimeters a year. How many centimeters in 30 years? How many meters is that?

4. **Engineering.** Explain one way engineers help buildings survive earthquakes and one way ShakeAlert helps people.

★ **Challenge: Find the mistake.** "Hawaii has volcanoes because it is on a plate boundary." Explain what is wrong, then sketch where Hawaii's volcanoes form.

Looking Ahead: Next week is the Unit 5 Science Checkpoint, reviewing Weeks 21 to 24: chemical changes, senses and signals, decomposers and moving plates.